

Reviewer Pack — Duality-Preserved Homotopy Stability

Entry 52b8188bdfa4 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Duality-Preserved Homotopy Stability

Entry ID: 52b8188bdfa4 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-01 03:54:57 UTC

1. Conjecture

Field A (mathematical branch): Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC) **Field B** (complexity object): Homotopy-stable phase spaces

Statement:

If a tropical circuit C satisfies A3's homotopy-stability condition, then its duality-flipped counterpart C' also exhibits homotopy-stable phase spaces under weight perturbations.

Rationale (proposer's reasoning):

A3 asserts that provable $\forall \Sigma_1^b$ sentences correspond to homotopy-stable phase spaces. Duality Flip preserves functional equivalence on integer inputs (per operations), so stability under perturbations should carry over to the dual circuit, maintaining the connection between proof validity and phase robustness.

Taxonomy category: BOUNDED_ARITHMETIC (status at proposal time:)

Framework membership: framework fw_a1a152ae17, role: elaboration

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 35ad90aecec86916

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.95	SAFE	SAFE
ALGEBRIZATION	SAFE	0.95	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropical circuit homotopy-stable phase space duality preservation - bounded arithmetic duality-flipped weight analysis homotopy stability - tropical circuit weight perturbations homotopy-stable phase space duality

Top relevant hits considered: - [<http://arxiv.org/abs/1207.2443v2>] Tropical Teichmuller and Siegel spaces - [<http://arxiv.org/abs/1204.6154v2>] Local Tropicalization - [<http://arxiv.org/abs/1204.3875v2>] Tropicalizing vs Compactifying the Torelli morphism - [<http://arxiv.org/abs/0706.3841v1>] Arithmetic lattices and weak spectral geometry - [<http://arxiv.org/abs/1512.00415v2>] Étale homological stability and arithmetic statistics - [<http://arxiv.org/abs/0906.0015v4>] On homotopy invariance for algebras over colored PROPs - [<http://arxiv.org/abs/1405.6940v3>] Tropical geometry of moduli spaces of weighted stable curves

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Define the tropical circuit and its dual
    def create_tropical_circuit(n):
        circuit = [[random.randint(-10, 10) for _ in range(n)] for _ in range(n)]
        return circuit

    def duality_flip(circuit):
        n = len(circuit)
        dual_circuit = [[circuit[j][i] for j in range(n)] for i in range(n)]
        return dual_circuit

    # Define homotopy stability check
    def is_homotopy_stable(circuit, epsilon=1e-6):
        n = len(circuit)
        original_solution = [sum(row[i] for row in circuit) for i in range(n)]
        perturbed_circuit = [[c + random.uniform(-epsilon, epsilon) if c != 0 else c for c in row]
                               for row in circuit]
        perturbed_solution = [sum(row[i] for row in perturbed_circuit) for i in range(n)]
        return all(abs(original_solution[i] - perturbed_solution[i]) <= epsilon for i in range(n))

    # Create a tropical circuit and its dual
    n = random.choice([5, 10, 15, 20, 30, 40])
    circuit = create_tropical_circuit(n)
    dual_circuit = duality_flip(circuit)

    # Check homotopy stability for both circuits
    original_stable = is_homotopy_stable(circuit)
    dual_stable = is_homotopy_stable(dual_circuit)

    return {
```

```

    "metric_name": "homotopy_stability",
    "metric_value": 1 if original_stable and dual_stable else 0,
    "instances_tested": 2,
    "conjecture_holds": original_stable and dual_stable,
    "counterexample": "" if original_stable and dual_stable else "Solution difference exceeds
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r['metric_value'] for r in results) / len(results)
    std_value = math.sqrt(sum((r['metric_value'] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r['conjecture_holds']) / len(results)

    if all(r['conjecture_holds'] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r['conjecture_holds'] for r in results):
        first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
        print(f"RESULT: FALSIFIED counterexample=\"Solution difference exceeds epsilon\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

e_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}
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TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ld1fe44e.py", line 73, in <module>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                        ~~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ld1fe44e.py", line 73, in <genexpr>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                        ~~~~~~

```

KeyError: 'seed'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed before completing required validation; counterexamples found but insufficient data to confirm falsification | next: Investigate crash cause in test_1d1fe44e.py's seed handling

11. Audit log (LLM calls)

Total LLM calls: 6

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	preregistration	ollama_remote	qwen3:8b	0	0	24367
2	novelty	ollama_remote	qwen3:8b	0	0	19580
3	novelty	ollama_remote	qwen3:8b	0	0	24692
4	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15835
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8594
6	judge	ollama_remote	qwen3:8b	0	0	13701

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 106770 ms total latency. Provider mix: {'ollama_remote': 6}

(full prompt+response transcripts available in research/audit/52b8188bdfa4.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/52b8188bdfa4.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/52b8188bdfa4.tar.gz (if generated)